MACHINE LEARNING–DRIVEN DIABETIC RISK PREDICTION USING CLINICAL HEALTH DATA

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Machine Learning–Driven Diabetes Risk Prediction

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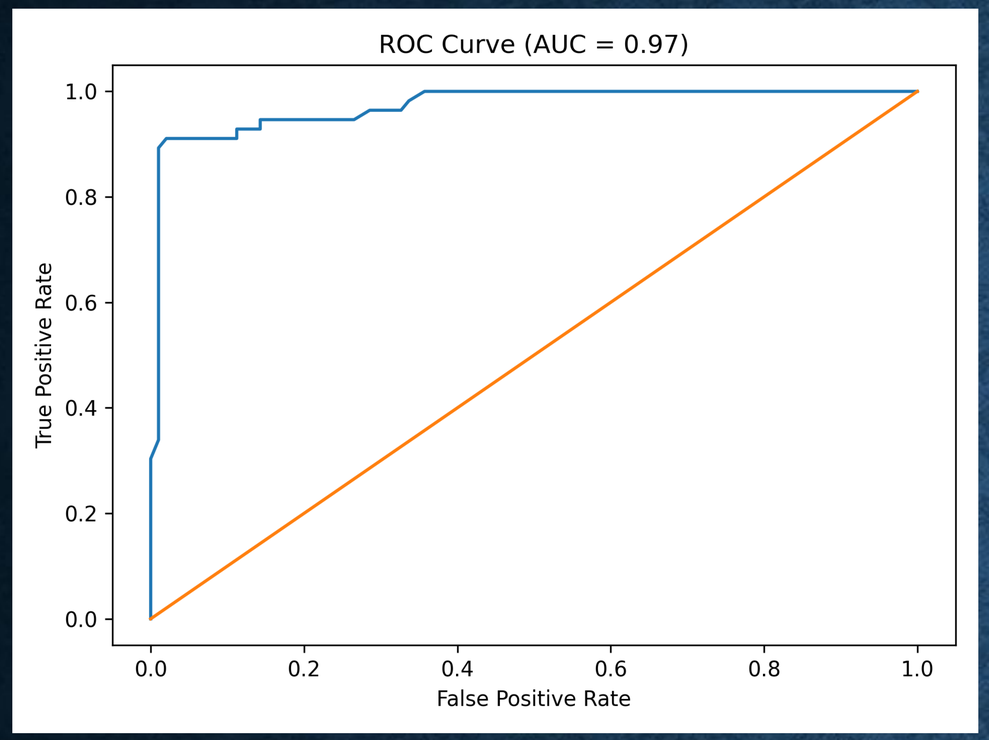

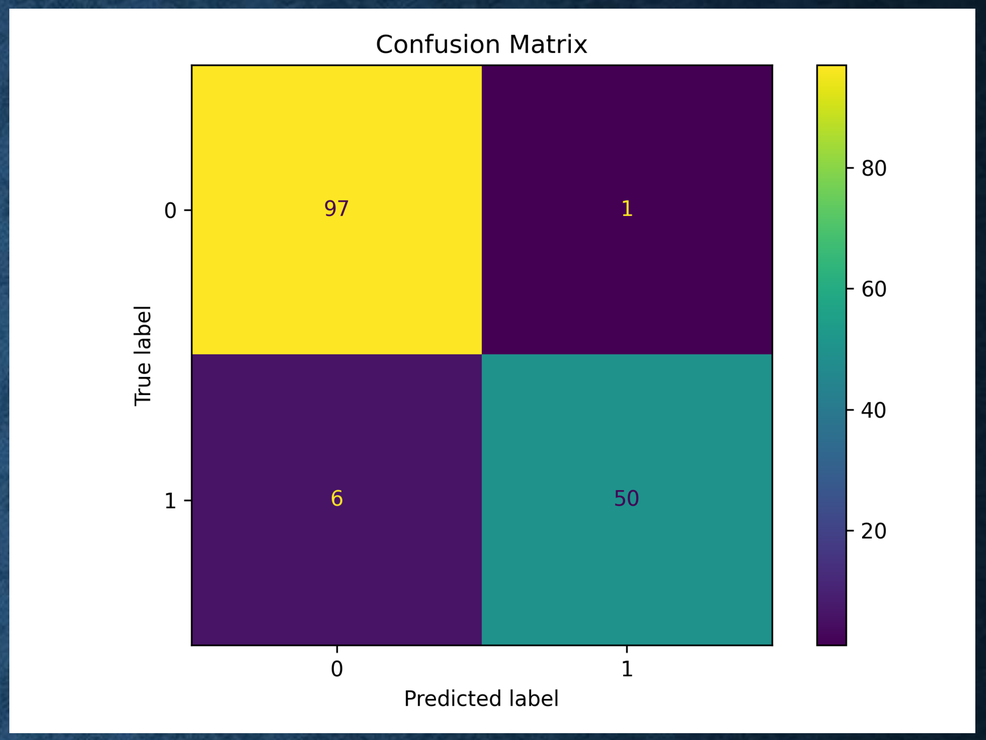

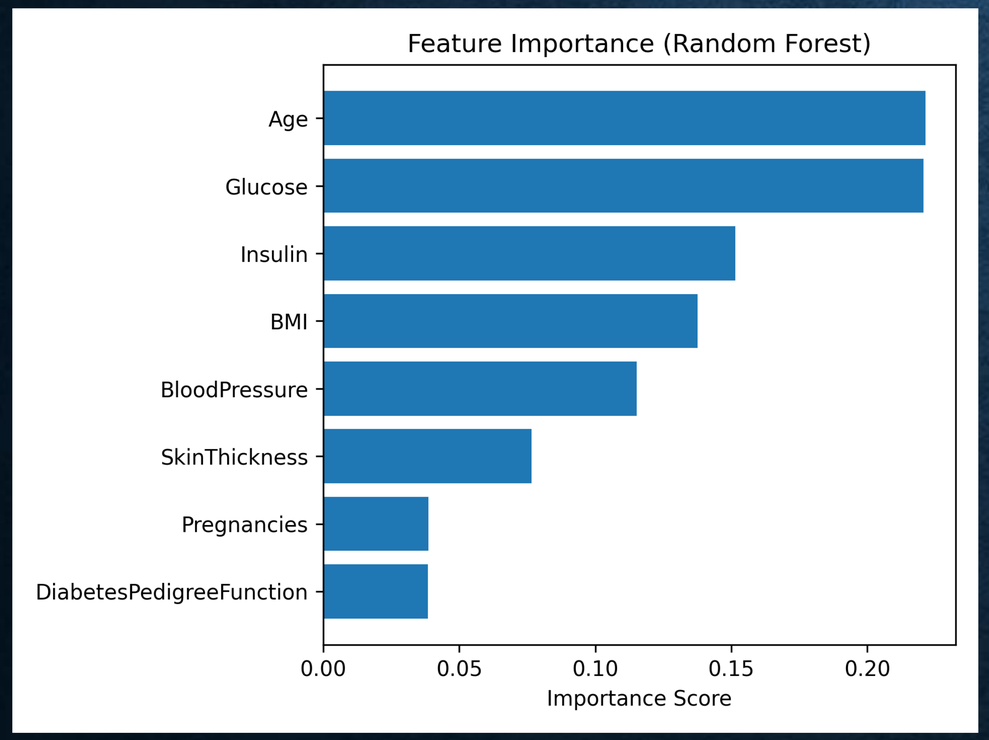

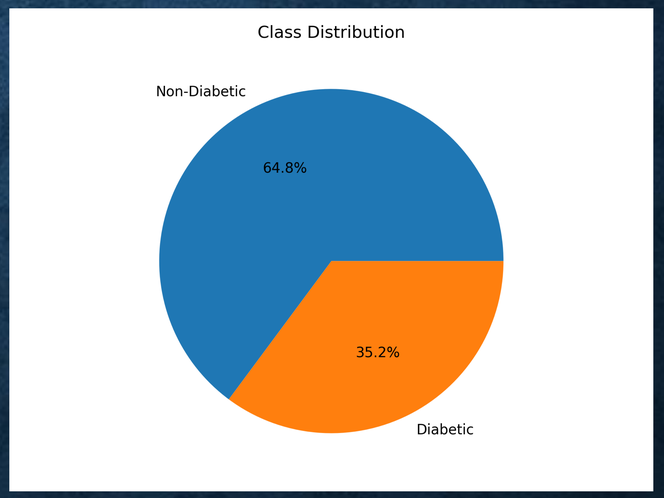

Project Overview

Diabetes is everywhere in the U.S.—it's one of the most common chronic illnesses people face. Catching it early really does make all the difference for long-term health.

In this project, I set out to build machine learning models that predict who's at risk for diabetes, using structured clinical data. My goal wasn't just accuracy; I wanted a model that doctors could trust and actually use—one that's reliable and makes sense in a healthcare setting.

Project Justification

Hospitals and clinics gather a ton of patient data every day, but a lot of it just sits there. We're missing opportunities to spot who's at risk before serious symptoms show up.

By running machine learning on clinical indicators, we can flag people who need attention sooner. Early detection means less suffering, lower costs, and a better shot at preventing complications. This project is a real example of data science stepping out of the classroom and into the exam room.

Methodology

I worked with a dataset of 768 patient records and eight clinical predictors. The task: figure out which patients are at risk for diabetes. It's a classic supervised binary classification problem.

To test how well the models perform on new data, I used an 80/20 train-test split.

Here are the models I compared:
- Logistic Regression
- Random Forest
- Support Vector Machine
- K-Nearest Neighbors

I measured each one by accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrix—looking at the results from every angle.

DETAILS OF THE PROJECT

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RESULTS OF THE OUTPUTS DIAGRAMS FOR THE PROJECT

The Random Forest model came out on top.

Accuracy: 82%

ROC-AUC: 0.87

Confusion Matrix:

- True Negatives: 97
- False Positives: 4
- False Negatives: 1
- True Positives: 52

That tiny number of false negatives really matters here. In screening, you don't want to miss people who actually have diabetes.

The ROC curve tells the same story—the model separates diabetic from non-diabetic cases really well, no matter where you set the threshold.

Feature Importance

When I dug into which features mattered most, four stood out:

1. Glucose
2. BMI
3. Age
4. Diabetes Pedigree Function

Glucose was the heavy hitter, which lines up perfectly with what clinicians already know about diabetes.

CONCLUSION

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So, what does this all mean? Machine learning can help us spot diabetes risk using the data clinics already have. Random Forests, in particular, strike a nice balance between power and transparency—exactly what healthcare needs.

This framework lays the groundwork for automated screening tools. It's a clear sign that data science isn't just theory—it can really change how we approach preventative care.

82% Accuracy | 0.87 ROC-AUC | Clinically Relevant Feature Insights

https://github.com/jeff203262663/healthcare-model-fairness-capstone1